

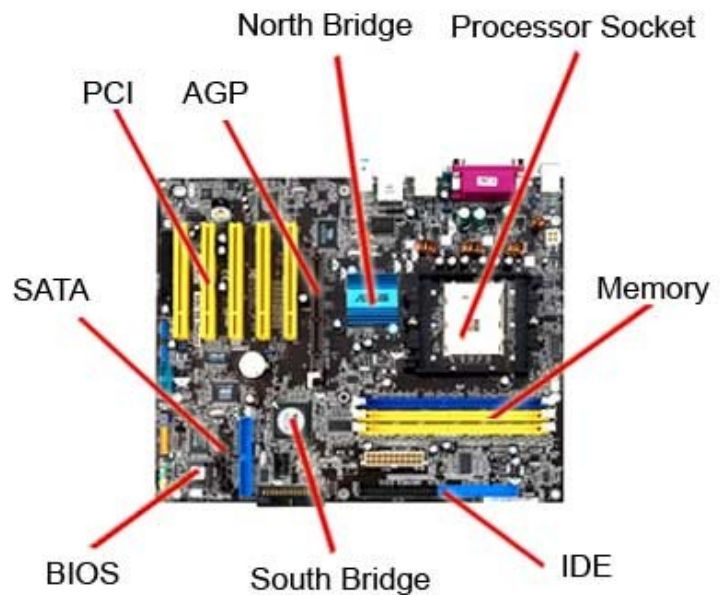
The Motherboard

A motherboard serves as the central hub that connects the hardware components of a computer, including the processor (CPU), memory (RAM), storage devices, and graphics card. Different types of motherboards are designed to accommodate various computer sizes and configurations.

Each motherboard is compatible with specific processors and memory types, meaning it cannot support every CPU or RAM module. However, most storage devices, such as hard drives, are largely universal and can work with a wide range of motherboards, regardless of their type or brand.

Components on Motherboard

- CPU socket.
- Northbridge.
- Southbridge.
- Memory slot.
- Expansion slots (PCI Express, PCI, and AGP).
- Coin cell battery (CMOS backup battery).
- BIOS
- Bus
- Slots

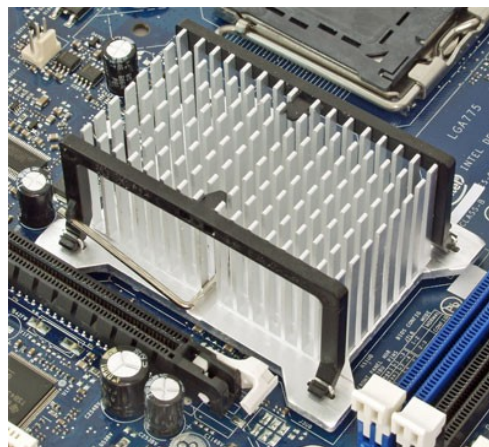


Northbridge

The Northbridge is an integrated circuit that facilitates communication between the CPU (Central Processing Unit), memory (RAM), and high-speed peripherals such as the GPU (Graphics Processing Unit) and expansion slots like PCIe (Peripheral Component Interconnect Express). It plays a critical role in managing data flow for these performance-intensive components. Unlike the Southbridge, which handles lower-speed peripherals and I/O functions, the Northbridge is directly connected to the CPU, RAM, and graphics controller and acts as a "bridge" for the Southbridge to communicate with these components.

Traditionally, both the Northbridge and Southbridge have dedicated heat sinks to manage heat dissipation due to their processing demands. The Northbridge is typically larger than the Southbridge and is located closer to the CPU and memory on the motherboard, reflecting its role in managing high-speed communication.

In modern computer architectures, much of the Northbridge's functionality has been integrated directly into the CPU or merged with the Southbridge to form a single chipset, often referred to as the Platform Controller Hub (PCH) in Intel systems. Despite this integration, the term "Northbridge" is still used informally to describe the component or functionality responsible for managing communication between the CPU, memory, and high-speed peripherals.



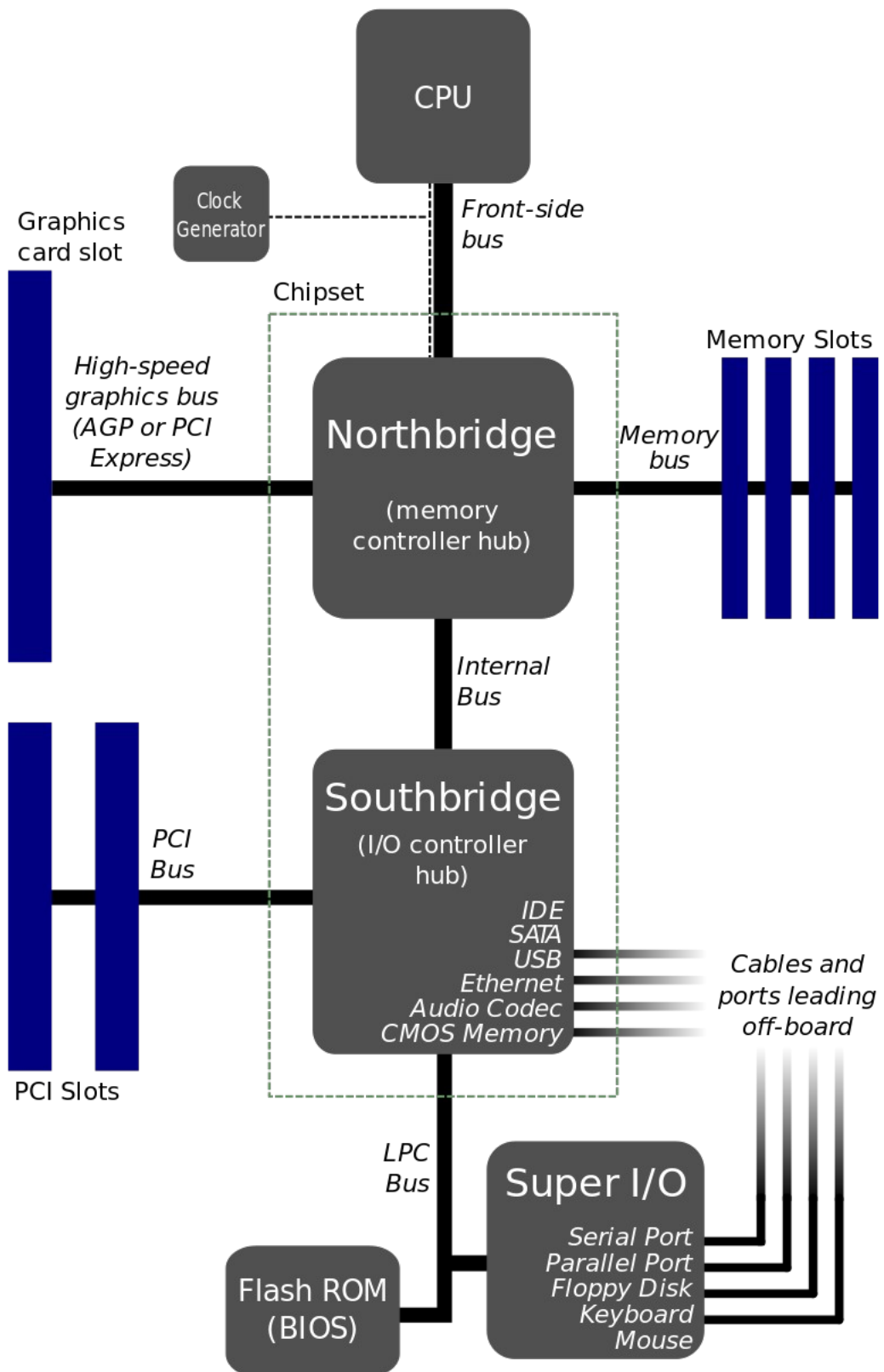
Southbridge

The Southbridge, also known as the I/O Controller Hub (ICH) or Platform Controller Hub (PCH) in modern systems, is an integrated circuit chip located on a computer's motherboard. It plays a vital role in managing input/output (I/O) functions and supporting lower-speed peripherals.

The Southbridge is responsible for coordinating communication between the CPU (via the Northbridge or directly in modern systems) and various I/O devices and interfaces, including:

- **Storage Devices:** Manages connections to storage devices such as hard disk drives (HDDs), solid-state drives (SSDs), optical drives (e.g., CD/DVD drives), and external storage devices.
- **Networking:** Handles network connectivity through Ethernet LAN (Local Area Network) controllers or other network interfaces.
- **Audio:** Facilitates audio functionality by managing audio inputs and outputs through integrated audio controllers.
- **USB (Universal Serial Bus):** Controls USB ports, enabling connections for peripherals like keyboards, mice, printers, and external drives.
- **Legacy Interfaces:** Supports legacy interfaces such as PS/2 ports for keyboards and mice, serial ports, parallel ports, and floppy disk drives (if present).
- **Power Management:** Provides power management features for system power efficiency, including support for sleep and hibernation modes.

While the Northbridge (or its integrated equivalent in modern CPUs) manages high-speed communication between the CPU, memory, and graphics subsystem, the Southbridge is focused on handling I/O functions and lower-speed peripherals. In modern architectures, the Southbridge's role has often been integrated into a single chip alongside Northbridge functions, forming a unified chipset like Intel's Platform Controller Hub (PCH).

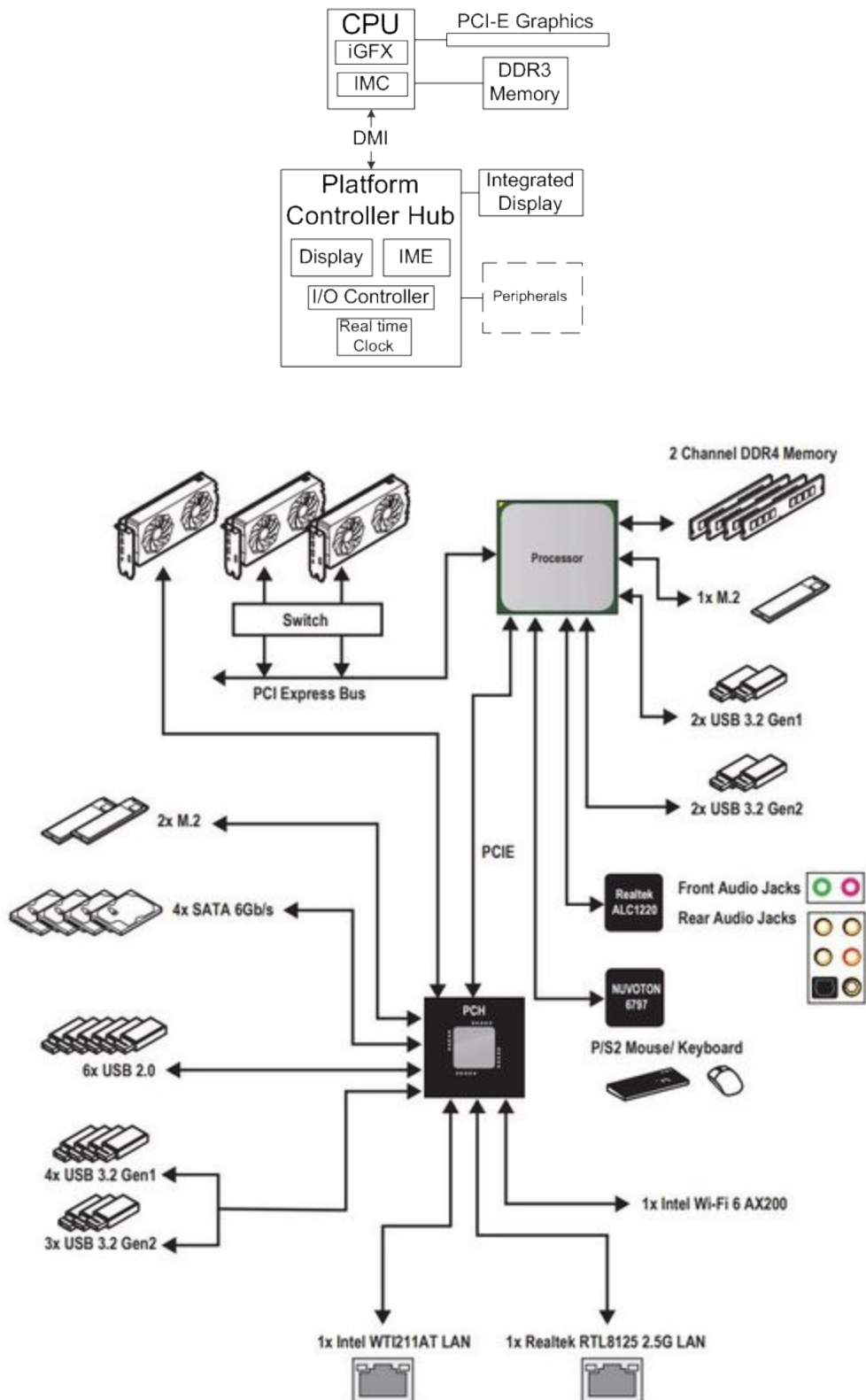


Platform Controller Hub

The Platform Controller Hub (PCH) is an integrated circuit chip located on a computer's motherboard. It serves as a central hub for managing various input/output (I/O) functions and system resources.

The PCH combines the functionalities of the traditional Northbridge and Southbridge chips into a single integrated circuit, facilitating a wide range of tasks:

- **I/O Interfaces:** Manages connections to storage devices (HDDs, SSDs), networking controllers (e.g., Ethernet LAN), USB ports, audio controllers, and legacy interfaces such as PS/2, serial, and parallel ports.
- **System Management:** Provides features for system and power management, including controlling system power consumption and supporting sleep, hibernation, and wake-on-LAN functionality.
- **Communication with the CPU:** Serves as a bridge between the CPU and various peripherals, enabling data exchange. The PCH communicates with the CPU via a high-speed interface, such as DMI (Direct Media Interface) in Intel systems.
- **Graphics Support:** While the PCH manages display output (e.g., HDMI or DisplayPort) by interfacing with the CPU's integrated graphics, the actual graphics processing is handled by the CPU, not the PCH.
- **Security Features:** Includes hardware-level security features such as secure boot, trusted platform module (TPM) support, and encryption capabilities to enhance system security.



BIOS (Basic Input/Output System)

The **BIOS** is firmware stored on a ROM (Read-Only Memory) chip on a computer's motherboard. It serves as a critical intermediary between the computer's hardware and its operating system (OS), performing key initialization tasks during the boot process.

Key Functions of the BIOS

1. Power-On Self-Test (POST):

- The BIOS conducts a series of diagnostic tests to ensure that critical hardware components, such as the CPU, memory (RAM), storage devices, and I/O devices, are functioning correctly.
- If errors are detected during POST, the BIOS generates error codes or audible beep patterns to indicate the problem.

2. Bootstrapping:

- After completing the POST successfully, the BIOS initiates the boot process by locating the operating system stored on a designated boot device (e.g., hard drive, SSD, or USB drive).
- It loads the boot loader program from the boot device into memory and transfers control to it, allowing the OS to take over the boot process.

3. Hardware Initialization and Configuration:

- The BIOS initializes and configures essential hardware components, including the CPU, memory, storage controllers, and peripheral devices.
- It sets default configurations to ensure compatibility and proper operation.

4. System Settings Configuration:

- The BIOS provides access to a configuration utility (commonly called the BIOS setup utility) that allows users to adjust system settings, such as boot device order, CPU performance settings, memory timings, and power management options.
- These settings are stored in a CMOS (Complementary Metal-Oxide-Semiconductor) chip and retained using a small battery on the motherboard.

5. Firmware Updates:

- The BIOS firmware can be updated (flashed) to fix bugs, add new features, or improve hardware compatibility.
- Updates are typically provided by the motherboard manufacturer and applied using tools provided by the manufacturer.

Here is the sequence of tasks performed from when the power button is pressed to when the operating system (OS) is loaded:

1. Power On

- When the power button is pressed, the power supply unit (PSU) provides electricity to the motherboard and other components.

2. Power Good Signal

- The PSU sends a "Power Good" signal to the motherboard, indicating that power is stable and the system can begin initialization.

3. CPU Initialization

- The CPU is activated first. It starts executing code stored in the firmware (BIOS/UEFI).

4. BIOS/UEFI Execution

- The BIOS/UEFI firmware is loaded and begins performing essential initialization tasks:
 - **Hardware Initialization:** Detects and initializes hardware components like RAM, storage drives, and input/output devices.

5. Power-On Self-Test (POST)

- The BIOS/UEFI performs a diagnostic test (POST) to ensure that critical hardware components (CPU, RAM, storage, GPU, etc.) are functioning properly.
- **Error Handling:** If any issues are detected, the system halts, and error codes or beep patterns are displayed.

6. System Configuration Load

- The BIOS/UEFI retrieves system settings stored in the CMOS memory (e.g., boot device priority, date, time) and applies them.

7. Boot Device Detection

- The BIOS/UEFI identifies the storage device configured as the primary boot device (e.g., SSD, HDD, or USB drive).

8. Bootloader Loading

- The BIOS/UEFI loads the bootloader (e.g., GRUB for Linux, Windows Boot Manager) from the boot device into memory.

9. Bootloader Execution

- The bootloader initializes and prepares the operating system for loading. It may present a menu for OS selection if multiple OSes are installed.

10. Operating System Kernel Load

- The bootloader loads the operating system kernel (the core part of the OS) into memory and transfers control to it.

11. OS Initialization

- The operating system initializes its core components:
 - Configures drivers for hardware.
 - Starts system services and background processes.
 - Initializes the user interface (e.g., desktop environment).

12. User Login

- **Description:** The OS displays the login screen, allowing the user to log in and begin using the system.

CMOS and Battery Backup

CMOS (Complementary Metal-Oxide-Semiconductor) refers to a semiconductor technology used in the construction of integrated circuits, particularly those that store small amounts of data. In the context of computing, **CMOS memory** refers to a type of volatile memory that is used to store system configuration settings, including BIOS settings, which are essential for booting and managing the system.

CMOS memory stores data such as:

- BIOS settings
- Date and time settings
- Boot device priority
- Hardware configurations

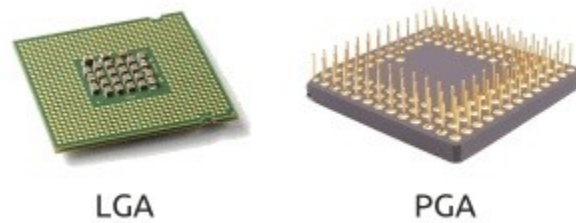
Unlike the main system memory (RAM), which loses its data when the power is turned off, CMOS memory retains its contents because it is continuously powered by a small, dedicated battery.

The **CMOS battery** is a small, round, coin-shaped battery, typically a **CR2032 lithium battery**, located on the motherboard. Its primary role is to provide backup power to the CMOS memory when the computer is powered off or unplugged, ensuring that the BIOS settings and configuration data are preserved.

If the CMOS battery becomes depleted or fails, the CMOS memory may lose its data. This can lead to issues such as:

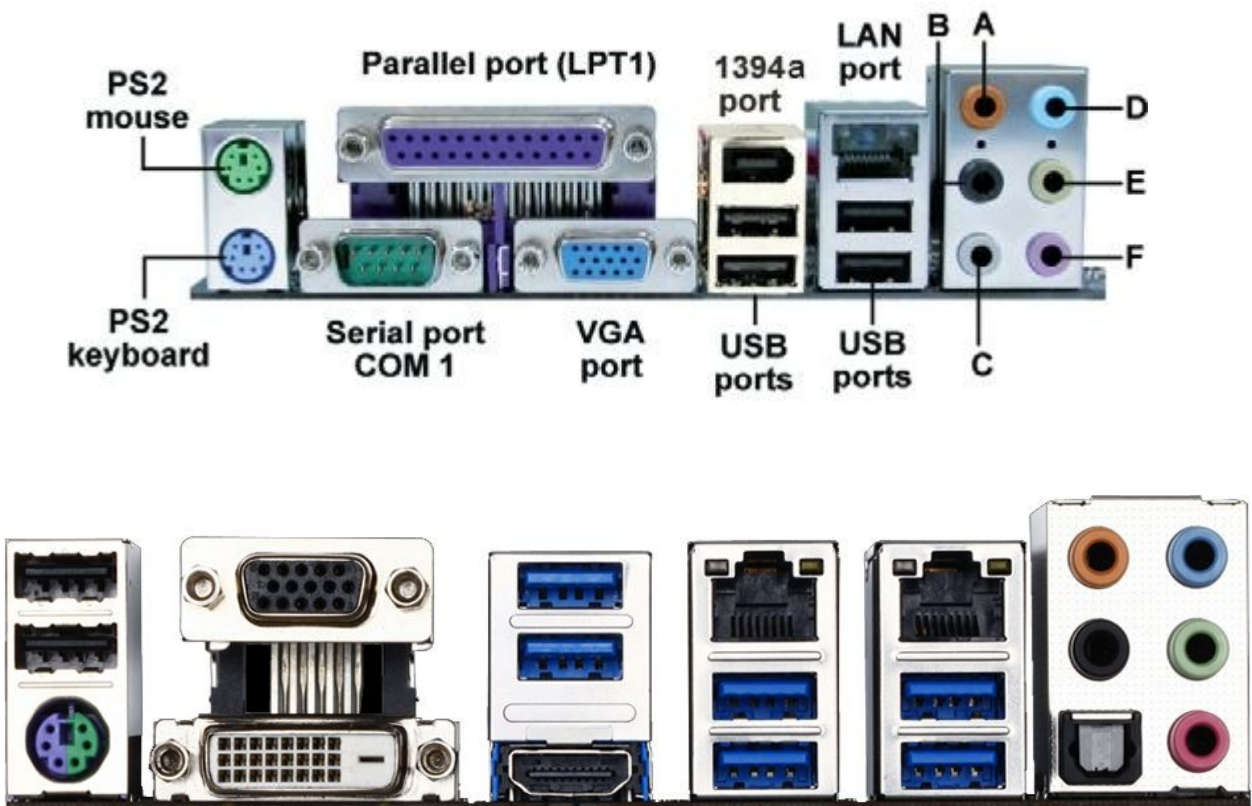
- Incorrect system date and time
- Loss of BIOS settings
- Boot failures or hardware detection problems

In such cases, replacing the CMOS battery with a new one (usually a **CR2032** lithium battery) will typically restore the BIOS settings and system functionality.

CPU sockets / Processor Interface

LGA - Land Grid Array where **pads** on the underside of the processor connect to pins in the processor socket on the motherboard.

PGA - Pin Grid Array where **pins** on the underside of the processor connect to holes in the processor socket on the motherboard.

Motherboard Back Panel Ports

USB (Universal Serial Bus)



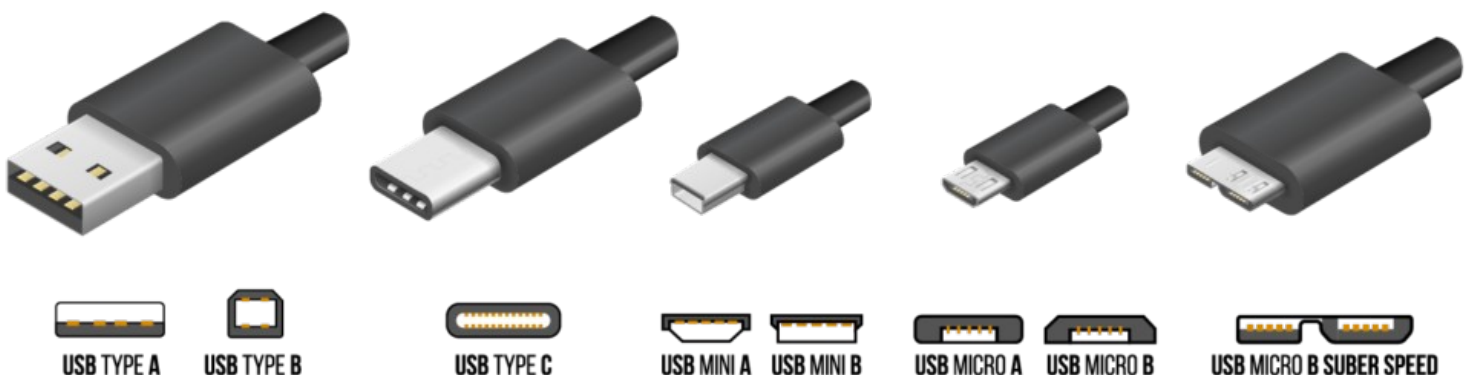
USB (Universal Serial Bus) is an industry standard that defines specifications for cables, connectors, and communication protocols used for connecting, communicating, and supplying power between computers, peripheral devices, and other electronic devices. First introduced in **1996**, the USB standard has significantly evolved over time, enabling faster data transfer speeds and improved power management.

The USB standard is maintained by the **USB Implementers Forum (USB-IF)**, a non-profit organization that oversees its development.

There have been several generations of USB specifications:

1. **USB 1.x**: Released in 1996, with **USB 1.1** being the most common version. It supports data transfer rates of up to **12 Mbps**.
2. **USB 2.0**: Introduced in 2000, offering a significant speed increase to **480 Mbps**.
3. **USB 3.x**: Released starting in 2008, with **USB 3.0** offering speeds up to **5 Gbps**. Later versions, including **USB 3.1** and **USB 3.2**, support speeds ranging from **5 Gbps** to **20 Gbps**.
4. **USB4**: Introduced in **2019**, offering speeds up to **40 Gbps** and better compatibility with Thunderbolt 3 devices.

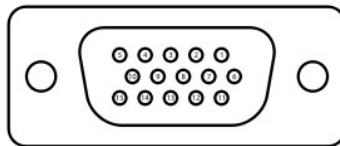
Each new generation of USB has introduced improvements in data transfer speeds, power delivery, and device compatibility.



VGA Port

VGA stands for **Video Graphics Array**, and it is an analog video connection standard used for connecting monitors or displays to computers. VGA is an older interface but still found on some legacy devices.

- **Signal Type:** VGA is an **analog** connection that transmits **RGBHV** (Red, Green, Blue, Horizontal Sync, and Vertical Sync) video signals.
- **Connector:** A standard VGA connector has **15 pins** and is typically shaped like a **trapezoid**.
- **Limitation:** VGA is limited by the fact that it is analog, meaning it is susceptible to signal degradation over longer cable lengths and doesn't support higher resolutions as well as digital connections.

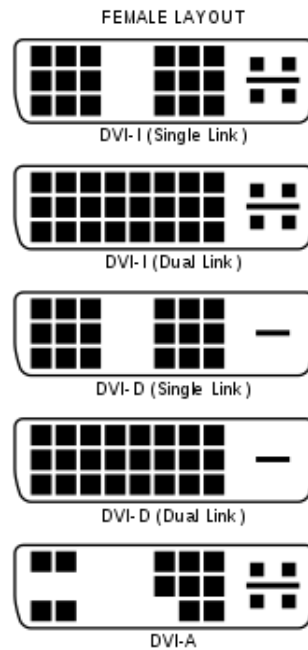


DVI Port (Digital Visual Interface)

DVI is a digital video connection standard designed to transmit both **analog and digital signals**. It supports higher resolutions and better image quality compared to VGA.

- **Signal Types:**
 - **DVI-I: Integrated** connector that supports both **digital and analog signals** (single or dual-link for digital).
 - **DVI-D: Digital-only** connector (can be single-link or dual-link).
 - **DVI-A: Analog-only** connector.
 -
- **Resolution and Bandwidth:** DVI supports high-bandwidth signals, with a maximum pixel clock over **160 MHz**, making it suitable for **high-resolution displays** such as **UXGA (1600x1200)** and **HDTV** resolutions.
- **Use Cases:** You may find DVI ports on video cards, monitors, and high-end televisions.

The digital video interface supports high bandwidth signals, over 160 MHz, which means it can be used for high resolution displays such as UXGA and HDTV. You may find DVI ports on video cards in computers as well as on high-end televisions.



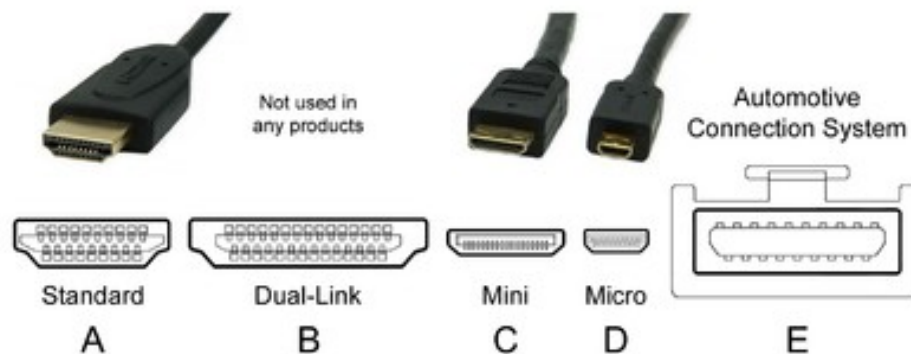
Additional Notes:

Conversion: If a display is analog and connected to a DVI-D or DVI-I port, the digital-to-analog conversion happens within the port (in the case of DVI-I). DVI-A connectors, however, transmit only analog signals and are primarily used for legacy analog displays.

HDMI (High-Definition Multimedia Interface)

HDMI (High-Definition Multimedia Interface) is a proprietary audio/video interface used for transmitting both **uncompressed video** and **compressed or uncompressed digital audio** from an HDMI-compliant source device (such as a computer, video game console, or Blu-ray player) to a compatible display or audio device (such as a television, computer monitor, video projector, or audio receiver).

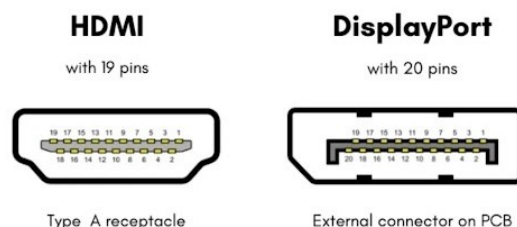
- **Video:** HDMI supports high-definition video signals, including **1080p**, **4K**, and higher resolutions, depending on the version of HDMI and the capabilities of the connected devices.
- **Audio:** HDMI can transmit multi-channel audio, including **stereo**, **5.1**, **7.1**, and lossless formats, such as **Dolby TrueHD** and **DTS-HD Master Audio**.
- **Compatibility:** HDMI is widely used in **consumer electronics** for high-quality digital video and audio transmission, supporting features like **HDCP (High-Bandwidth Digital Content Protection)** to prevent unauthorized copying of digital content.



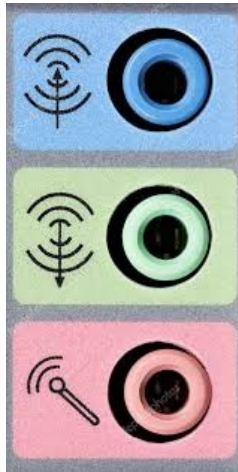
Display Port

DisplayPort is a **digital display interface** primarily used for connecting a computer to a monitor or other display device. It was developed by the **Video Electronics Standards Association (VESA)** as a successor to older interfaces like **VGA** and **DVI**, offering **higher bandwidth** and more advanced features.

- **Video and Audio:** DisplayPort supports both **high-definition video** and **digital audio** signals. It supports **high resolutions**, such as **4K**, **8K**, and beyond, as well as high **refresh rates** for smooth, fluid motion.
- **Multiple Monitors:** DisplayPort has the ability to drive **multiple monitors** from a single connection using features like **Multi-Stream Transport (MST)** hubs and **daisy-chaining**. This allows users to connect several monitors through one port.
- **Single Cable:** One of the key features of DisplayPort is its ability to transmit both video and audio signals over a single cable, eliminating the need for separate audio cables. This is particularly useful when connecting computers to monitors that have **built-in speakers** or for use with **external audio systems**.
- **Compatibility:** DisplayPort is commonly used for **high-performance monitors** and supports higher bandwidths than HDMI in certain versions. It also has an alternative connector, **Mini DisplayPort**, often found in devices like laptops and ultrabooks.



Sound Port



Motherboard form factors and types

As computers advanced, so have motherboards. Below is a list of the various motherboard form factors and additional information about each, including ATX, which is the most common.

- AT
- ATX
- Baby AT
- BTX
- DTX
- LPX
- Full AT
- Full ATX



Standard-ATX



Micro-ATX



Mini-ITX

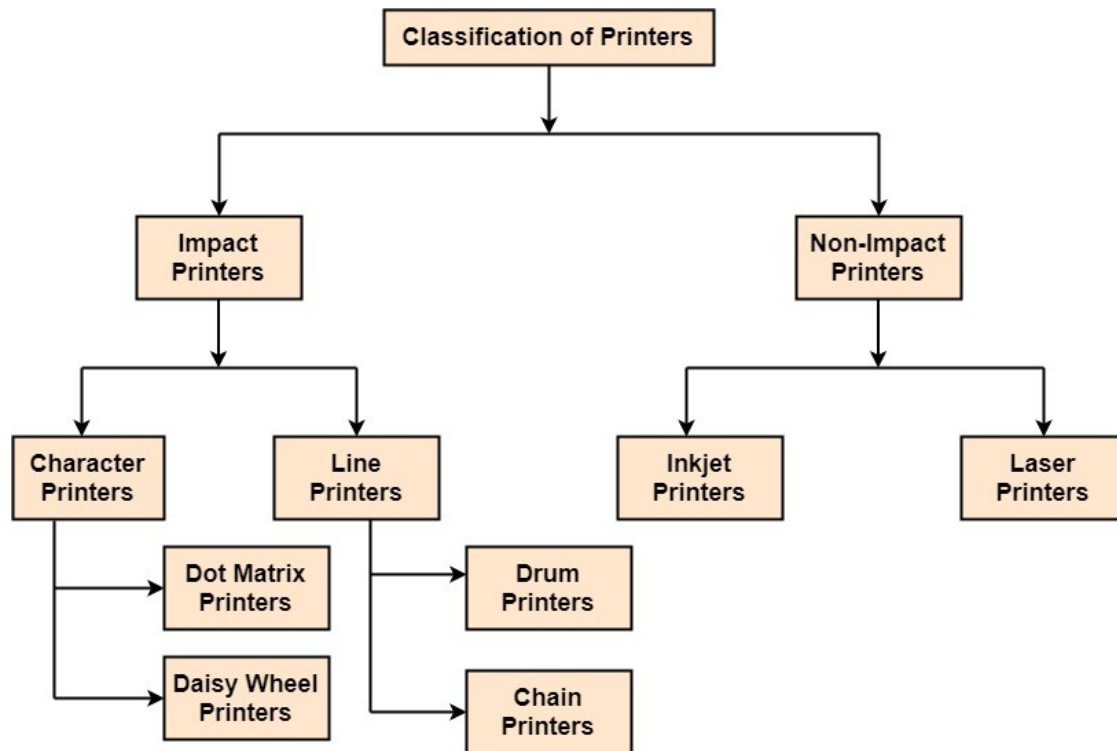


Nano-ITX



Pico-ITX

Printers



Impact Printers

An **impact printer** creates text or images on paper by physically striking the paper with an inked ribbon or similar medium using a metal, plastic, or rubber tool. Examples of impact printers include **typewriters**, **dot matrix printers**, **letterpress printers**, and **rubber stamps**.

- **Typewriter:** Creates text by striking the paper with an inked ribbon.
- **Dot Matrix Printer:** Uses a series of pins to form characters and images by striking an inked ribbon onto the paper.
- **Letterpress:** A traditional printing method where a raised image or text is pressed onto paper.

Non-impact printers

A **non-impact printer** creates an image or text on paper without directly striking the surface. Instead, it uses techniques like spraying ink, electrostatic transfer, or offset printing to produce the image. Examples of non-impact printers include **inkjet printers**, **laser printers**, and **copiers**.

- **Inkjet Printer:** Sprays tiny droplets of ink onto the paper to form images or text.
- **Laser Printer:** Uses electrostatic charges to transfer toner to the paper, which is then fused using heat.

Character printer

Character printers print one character at a time. They store individual characters and print them as needed. While they are slower compared to more modern printers, they were useful in environments where only text needed to be printed. Examples include **typewriters** and **some older printers**. They are rarely used today due to their slow speed and limited functionality.

Line Printers

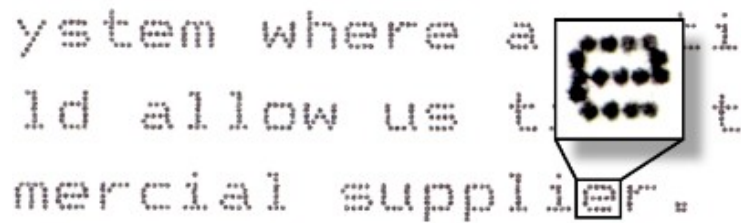
Line printers print one entire line of text at a time, making them much faster than character printers. These printers are commonly used in environments where high-volume printing is required. **Line-at-a-time printers** include **drum printers**, **chain printers**, and **band printers**.

- **Drum Printers:** Use a rotating drum with characters imprinted on it to print one line at a time.
- **Chain Printers:** Use a rotating chain with characters that print one line at a time.
- **Band Printers:** Similar to chain printers but use a band of characters.

Dot-Matrix Printers

Dot-matrix printers use a print head with an array of pins (ranging from 9 to 24 pins) to strike an inked ribbon against paper, forming characters and images by creating patterns of dots. A **24-pin dot-matrix printer** produces higher quality and more detailed characters compared to a **9-pin printer**.

- The print head moves across the paper in both directions (left to right, and right to left).
- Color output can be achieved by replacing the black ribbon with one that contains colored stripes.



Daisy-Wheel Printers

Daisy-wheel printers are impact printers that are used to achieve the high-quality text printing similar to that produced by typewriters. They are called **daisy-wheel printers** because the print mechanism resembles a daisy, with each "petal" containing a fully formed character. When printing, a hammer strikes the "petal" containing the character, pressing it onto the ribbon and then onto the paper to create the printed text.

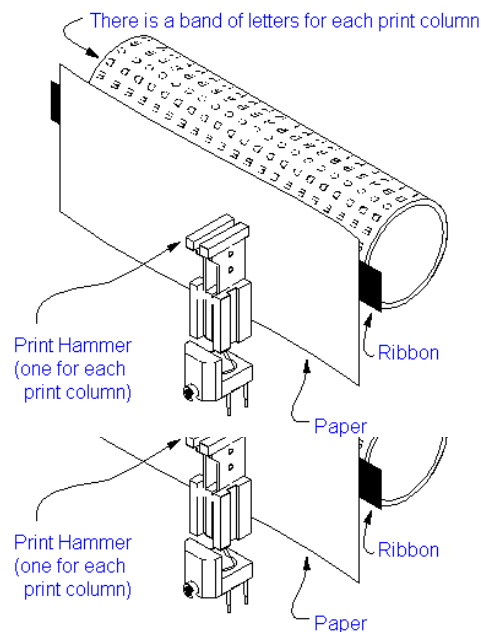
- The print head moves to the correct character on the "daisy" and strikes it against the inked ribbon to transfer the character onto the paper.
- Daisy-wheel printers are capable of producing solid, clear text, similar to the quality of typewriters.
- The printing speed of daisy-wheel printers typically ranges from **25 to 55 characters per second**, which is slower compared to more modern printers.



Drum Printer

A drum printer consists of a solid, cylindrical drum that has raised characters in bands on its surface. The number of print positions across the drum equals the number available on the page, typically ranging from 80 to 132 print positions. The drum rotates at a rapid speed, with a print hammer located behind the paper for each possible print position. These hammers strike the paper, along with the ink ribbon, against the proper character on the drum as it rotates. One revolution of the drum is required to print each line.

While all characters on the line are not printed simultaneously, the speed at which the entire line is printed is fast enough to classify them as line printers. Typical speeds of drum printers range from 300 to 2,000 lines per minute.

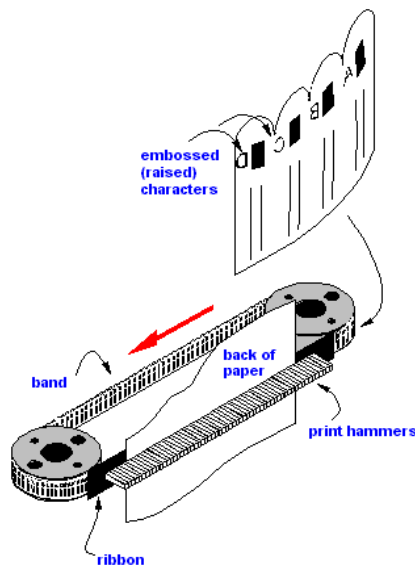


Chain Printers

A **chain printer** uses a continuous chain of print characters wrapped around two pulleys. Like the drum printer, there is one hammer for each print position. The printer's circuitry detects when the correct character on the chain reaches the desired print position on the page.

- When the correct character aligns with the print position, the hammer strikes the paper, pressing it against the ribbon and the character on the chain, transferring an impression of the character to the page.
- The chain keeps rotating until all the required print positions for the line are filled.
- After one line is printed, the paper moves up to print the next line.

Chain printers are known for their speed, with typical printing rates ranging from **400 to 2,500 characters per minute**.



Thermal printers

Thermal printers operate by selectively heating regions of heat-sensitive paper to produce text or images. The heat causes a chemical reaction in the paper, which results in a visible mark.

These printers are commonly used in devices like **cash registers**, **ATMs**, **gasoline dispensers**, and **some fax machines**. Thermal printers can produce high-quality prints quickly and are widely used for printing receipts and labels.

- **Color Printing:** Thermal printers can achieve different colors by using special **thermal papers** that respond to various temperatures. By controlling the heating rates and temperatures, the printer can produce multiple colors, though this is more commonly seen in **color thermal transfer printing**. This involves a thermal transfer ribbon instead of direct heating on the paper.



Ink-Jet Printers

Inkjet printers create images and text by spraying tiny droplets of ink onto paper. These droplets are emitted from small nozzles in the print head, which is controlled by an electrical field.

- The ink droplets are electrically charged after leaving the nozzle, and deflection plates guide the droplets to the correct position on the paper to form characters or images.
- The ink is absorbed by the paper and dries quickly, allowing for fast printing.

Color Printing: Inkjet printers typically use four primary colors: **Cyan, Magenta, Yellow, and Black** (CMYK). By mixing these colors in various proportions, inkjet printers can produce a wide range of colors.



Laser Printer

Laser printers create images on paper by directing a laser beam onto a mirror, which reflects the beam onto a photosensitive drum. The drum's surface is coated with a special material, typically made of selenium or organic photoconductors, that retains an electrostatic charge.

When the laser beam strikes the drum, it selectively discharges areas corresponding to the image being printed, forming a pattern of small dots. Toner, a fine powder containing pigments, is attracted to the discharged areas on the drum, adhering to its surface.

As the paper moves past the drum, the toner is transferred onto the paper, forming the desired text or graphics. To permanently bond the toner to the paper, the paper passes through a pair of heated rollers, known as the fuser assembly, which melts the toner and causes it to adhere to the paper.

Advantages:

- **Speed and efficiency:** Laser printers print quickly and efficiently, especially for high-volume tasks.
- **High-quality output:** They produce high-quality graphics and text, especially for detailed documents.
- **Quiet operation:** Laser printers are relatively quiet compared to other printer types.
- **Color printing capability:** Many laser printers are capable of printing in color.

Disadvantages:

- **Cost:** Laser printers are relatively expensive compared to inkjet or other printer types.
- **Maintenance and repair costs:** They can have high maintenance, repair, and servicing costs.
- **Health and environmental concerns:** Laser printers emit small amounts of ozone, which can be hazardous to health and the environment.

Graphic Plotter

A **graphic plotter** is a specialized output device used to produce high-quality, large-scale graphics, such as engineering drawings, architectural blueprints, and maps. Unlike standard printers, which typically print text and images on paper, graphic plotters use pens, pencils, or other drawing tools to physically draw lines and shapes on paper or other materials.

Graphic plotters are commonly used in fields such as engineering, architecture, cartography, and other technical industries where precision and detailed drawings are required. These devices are capable of producing intricate designs with high accuracy and consistency.

The plotter operates by moving the pen or drawing tool across the surface of the paper in two dimensions, with precise control over its position and movement. Some advanced graphic plotters may also rotate or adjust the angle of the drawing tool to create more complex shapes and patterns.

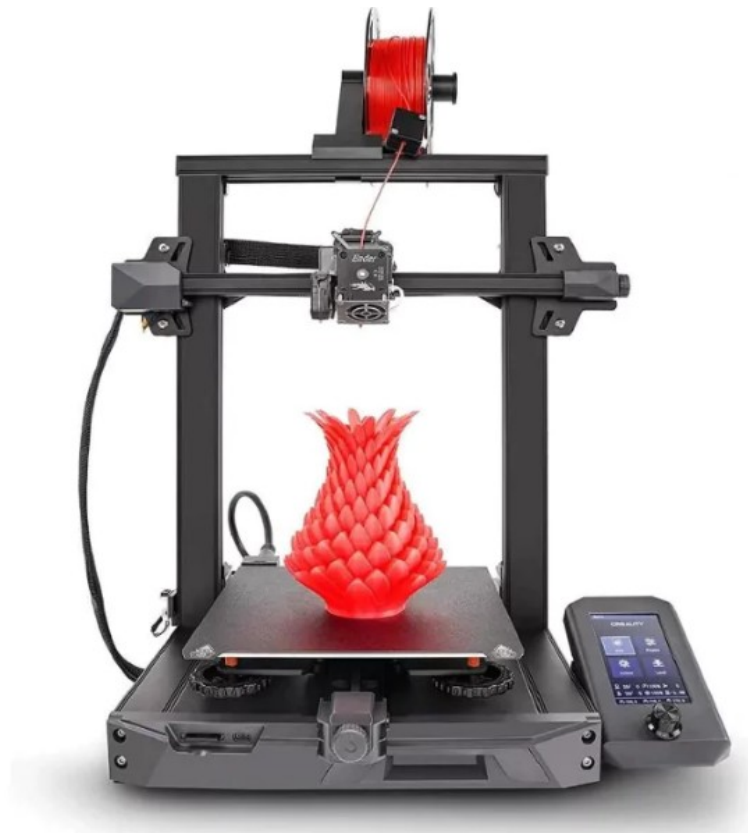


3D Printer

A **3D printer** is an additive manufacturing technology that creates three-dimensional objects layer by layer from a digital model. Unlike traditional subtractive manufacturing, which removes material from a solid block, 3D printing builds objects by depositing material layer by layer to form the desired shape.

These printers use a variety of materials, including plastics, metals, ceramics, and even biological materials, to create objects with complex geometries and fine details. The process begins with a digital 3D model of the object, which is then sliced into thin horizontal layers using specialized software. The 3D printer follows these instructions to deposit or solidify material one layer at a time, gradually building the final object.

Applications: 3D printing is used in numerous industries, including prototyping, product development, manufacturing, healthcare, aerospace, automotive, architecture, and education. It enables rapid, cost-effective production of custom parts, prototypes, tools, and even finished products, making it a versatile and innovative manufacturing method.



Properties of Printers

1. Resolution

DPI (dots per inch) measures the output resolution of a printer, indicating how many dots are present in a linear inch. Higher DPI results in finer detail in printed images. Printers with higher DPI are typically able to produce more detailed images, making them ideal for high-quality photo printing.

2. Printing Speed

Printing speed was historically measured differently depending on the type of printer:

- **Character printers:** Measured in **characters per minute (cpm)**.
- **Line printers:** Measured in **lines per minute (lpm)**.
- **Page printers:** Now commonly measured in **pages per minute (ppm)**.

However, these measurements are often used for marketing purposes and may not always be standardized. The **ppm** rate typically applies to sparse monochrome office documents and does not account for dense color prints or graphics-heavy documents. Speeds are usually specified for **A4** paper in most regions.

3. Cost

This refers to the initial purchase cost of the printer, as well as ongoing costs such as ink or toner, paper, and maintenance. Cost can vary widely depending on the type of printer (inkjet, laser, 3D, etc.).

4. Size

The physical dimensions of the printer can vary. Larger printers can handle higher volumes of printing, but they require more space and are usually more expensive. Size also impacts portability, especially for home or office use.

5. Color Printing Capability

Printers can be categorized as monochrome (black-and-white) or color printers. **Color printing capability** is a critical feature for printing high-quality graphics, photos, or documents that require multiple colors. Some printers can only print in a limited color range, while others, like inkjet printers, are capable of full color printing (typically using CMYK).

6. **Media Handling**

This refers to the types and sizes of paper or other materials the printer can handle. Printers can support various sizes (e.g., letter, legal, A4, A3), paper types (e.g., glossy, matte, card stock), and other media like labels or transparencies.

7. **Energy Consumption**

Printers, especially laser printers, can consume a significant amount of energy. This property is important when considering the environmental impact of printing, as energy-efficient printers can lower operating costs and reduce energy use over time.